Lab 3
IR Sensor

Course: RoboLink: Build and Drive Your Smartphone-Controlled Robot
Program: Mini Course Program 2026, Carleton University
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Lab Mission
In this lab, you will use an IR obstacle sensor to detect an object. The Arduino will read the sensor, turn on an LED, and make a buzzer sound when something is detected. This is one of the first steps toward building a robot that can sense its surroundings.

Group Member Names ______________________________

Date ______________________________

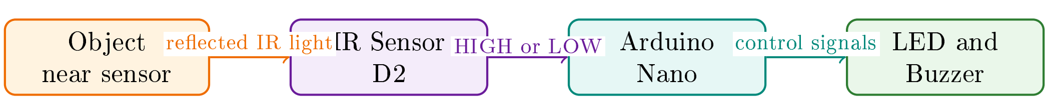
1. Learning Goals

By the end of this lab, you should be able to:

- Identify the main pins on an IR obstacle sensor module.
- Wire an IR sensor, LED, and buzzer to an Arduino Nano ATmega168.
- Read a digital sensor value using `digitalRead()`.
- Use an `if/else` decision to control outputs.
- Explain why many IR sensors read `LOW` when an object is detected.
- Use the Serial Monitor to test whether the sensor is reading correctly.
- Troubleshoot common wiring and sensor-detection problems.

2. Big Idea

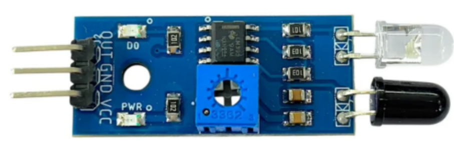
A sensor turns a real-world change into a signal.
The IR sensor checks whether reflected infrared light comes back from an object. The Arduino reads the sensor output pin as `HIGH` or `LOW`. Then the Arduino decides whether to turn on the LED and buzzer.



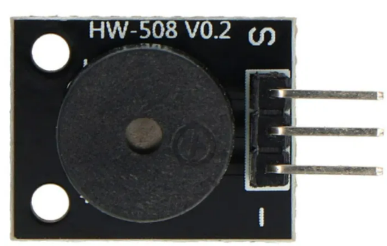
3. Materials

Arduino Nano ATmega168	Reads the IR sensor and controls the outputs.
Computer with Arduino IDE	Uploads the Lab 3 sketch and opens the Serial Monitor.
USB cable	Connects the Arduino Nano to the computer.
IR obstacle sensor module	Detects nearby objects using infrared light.
Three-pin buzzer module	Makes sound when an object is detected.
LED and resistor	Gives a visual warning when the sensor detects an object.
Breadboard and jumper wires	Connect the circuit without soldering.

4. Meet the Parts



IR obstacle sensor. The pins are usually labelled VCC, GND, and OUT. Some modules also include an analog output pin.



Buzzer module. The signal pin connects to the Arduino output pin.

IR Sensor Logic
Many IR obstacle sensors are **active-low**. This means the sensor output becomes `LOW` when it detects an object. If your sensor works the opposite way, the code can be changed to check for `HIGH` instead.

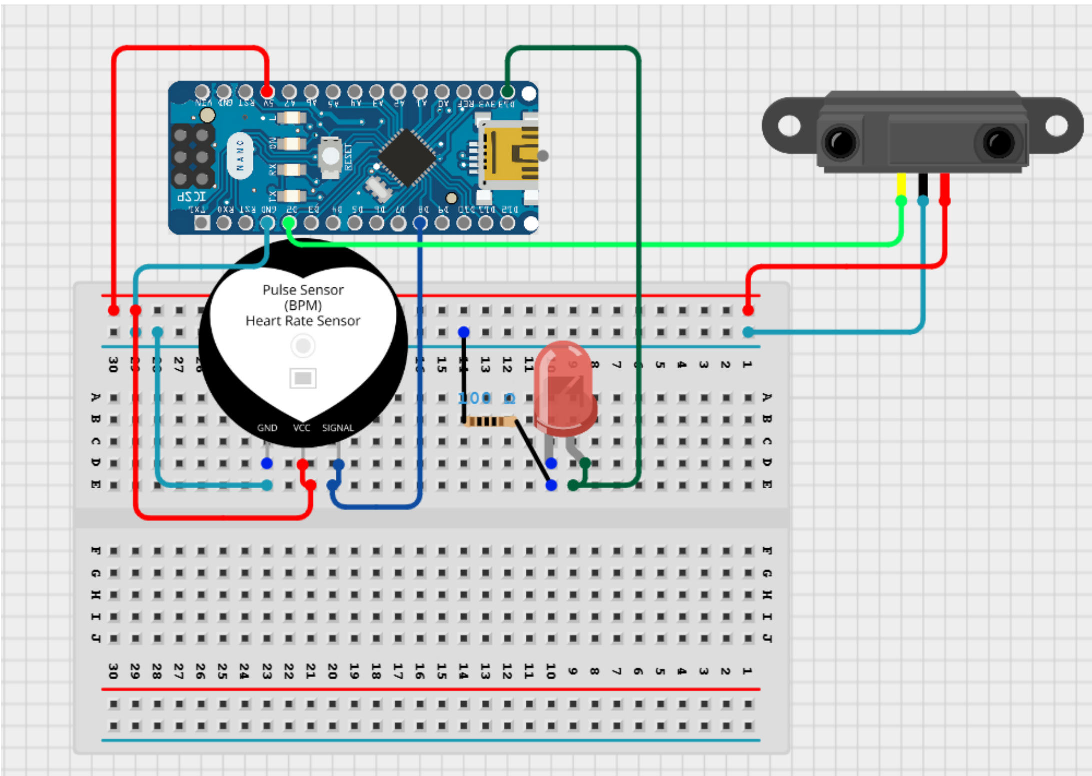
5. Safety and Setup Notes

Check Wiring Before Power
Ask your teacher to check the circuit before plugging in the Arduino. A wrong connection can stop the sensor from working or can damage a part.

Use a Resistor With the LED
If you are using an external LED, connect a resistor in series with it. A common value is 220 ohms to 330 ohms.

Use the Serial Monitor
The code prints the sensor value. Open the Serial Monitor at **9600 baud** so you can see whether the sensor is reading `0` or `1`.

6. Circuit Wiring



Lab 3 circuit. The IR sensor controls the LED and buzzer.

Part	Arduino Pin	Connection Notes
IR sensor OUT	D2	The Arduino reads this digital signal.
IR sensor VCC	5V	Powers the sensor module.
IR sensor GND	GND	Shares ground with the Arduino.
Buzzer signal	D8	The Arduino sends the sound signal here.
Buzzer VCC	5V	Powers the buzzer module.
Buzzer GND	GND	Shares ground with the Arduino.
LED positive leg	D13	Use a resistor if using an external LED.
LED negative leg	GND	Connects the LED back to ground.

7. Open the Code

Open this sketch in the Arduino IDE:

`Day3_Sensors/Labs/IR_with_Buzzer/IR_with_Buzzer.ino`

Teacher checkpoint: Before uploading, show your teacher that your board is set to **Arduino Nano**, your processor is set to **ATmega168**, and the correct port is selected.

8. Important Code Pieces

Pin Setup

```
const byte IR_SENSOR_PIN = 2;
const byte BUZZER_PIN = 8;
const byte LED_PIN = 13;
const unsigned int BUZZER_FREQUENCY = 1000;
```

`IR_SENSOR_PIN`	The Arduino pin that reads the IR sensor output.
`BUZZER_PIN`	The Arduino pin that controls the buzzer.
`LED_PIN`	The Arduino pin that controls the LED.
`BUZZER_FREQUENCY`	The sound pitch used by the buzzer, measured in hertz.

Setup

```
void setup() {
  Serial.begin(9600);
  pinMode(IR_SENSOR_PIN, INPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  pinMode(LED_PIN, OUTPUT);
}
```

Sensor Decision

```
sensorStatus = digitalRead(IR_SENSOR_PIN);

if (sensorStatus == LOW) {
  digitalWrite(LED_PIN, HIGH);
  tone(BUZZER_PIN, BUZZER_FREQUENCY);
} else {
  digitalWrite(LED_PIN, LOW);
  noTone(BUZZER_PIN);
}
```

What is the code deciding?
The Arduino asks one question: “Did the IR sensor detect an object?” If the answer is yes, it turns on the warning outputs. If the answer is no, it turns them off.

9. Experiment Procedure

Part A: Build and Upload

1. Build the circuit using the wiring table.
2. Ask your teacher to check the sensor, buzzer, LED, 5V, and GND wiring.
3. Open `IR_with_Buzzer.ino`.
4. Select **Arduino Nano**, **ATmega168**, and the correct port.
5. Click **Verify**. If there are no errors, click **Upload**.
6. Open the Serial Monitor and set it to **9600 baud**.

Part B: Test the Sensor

1. Keep your hand far away from the IR sensor.
2. Record the value printed in the Serial Monitor.
3. Move your hand slowly in front of the IR sensor.
4. Record the new value printed in the Serial Monitor.
5. Watch the LED and listen to the buzzer.

Test	Serial Monitor Value	LED and Buzzer Observation
No object near sensor		
Hand near sensor		
Object far away		
Object very close		

Part C: Adjust and Explore

Some IR sensor modules have a small adjustment screw. This changes the detection distance.

1. Ask your teacher before adjusting the screw.
2. Turn it only a small amount.
3. Test again with your hand at different distances.
4. Record the farthest distance where the sensor still detects your hand.

Maximum detection distance:

10. Mini Challenge

Choose one small code change. Upload again and test your circuit.

- Change `BUZZER_FREQUENCY` from `1000` to `500`.
- Change `BUZZER_FREQUENCY` from `1000` to `2000`.
- Change `delay(1000)` to `delay(250)`.
- Reverse the decision only if your sensor reads `HIGH` when it detects an object.

Challenge result: What did you change, and what happened?

11. Troubleshooting

Problem	Possible Cause	What To Try
Upload fails	Wrong board, processor, or port	Select Arduino Nano, ATmega168, and the correct port.
Serial Monitor shows nothing	Wrong baud rate or Serial Monitor not open	Set Serial Monitor to 9600 baud.
Sensor value stuck	OUT pin, 5V, or GND may be wrong	Check D2, 5V, and GND wiring.
LED stays off	LED direction or D13 wiring may be wrong	Check the LED legs, resistor, D13, and GND.
Buzzer does not beep	Signal, VCC, or GND may be wrong	Check D8, 5V, and GND.
LED and buzzer turn on at the wrong time	Sensor logic may be opposite	Change the `if` check from `LOW` to `HIGH`.
Detection distance is too short	Sensor adjustment or object surface may affect detection	Adjust the sensor gently and try a different object.

12. Reflection Questions

Answer in full sentences.

1. What does an IR obstacle sensor detect?

2. What value did your sensor print when no object was detected?

3. What value did your sensor print when an object was detected?

4. Why does the Arduino use an `if/else` statement in this lab?

5. Why is the buzzer connected to an output pin?

6. What could a robot do if it detects an object in front of it?

7. What was one problem your group solved during this lab?

13. Exit Ticket

In one or two sentences: Explain how the IR sensor makes the Arduino turn on the buzzer.

14. Teacher Check

Teacher Checklist

- [ ] **IR sensor wiring checked** Notes: ______________
- [ ] **Buzzer wiring checked** Notes: ______________
- [ ] **LED wiring checked** Notes: ______________
- [ ] **Arduino Nano and ATmega168 selected** Notes: ______________
- [ ] **Code uploaded** Notes: ______________
- [ ] **Serial Monitor tested at 9600 baud** Notes: ______________
- [ ] **Sensor detects object correctly** Notes: ______________
- [ ] **Students completed observations and reflections** Notes: ______________